TITAN: Analytical FMM/M2L Boundary-Integral Poisson-Boltzmann Solves with Sparse Near-Field ILU(0) Preconditioning

Submission-polished evidence-backed draft

Abstract

Grid Poisson-Boltzmann solvers remain robust workhorses, but their volumetric discretization can make memory, boundary smoothing, and interface resolution the dominant numerical compromises. Boundary-integral formulations remove the volume grid by solving directly on the molecular dielectric interface, yet they replace the grid problem with a dense derivative-rich surface operator and a stiff Krylov system. We present a guarded fourth-order analytical FMM/M2L path in TITAN that combines analytic-singular near-field integration, deterministic far-cell refinement, compressed Green-function jets, packed AVX2 direct fallback, and sparse near-field ILU(0) preconditioning. The method is evaluated as an advanced explicit TABI treecode option, not as a default backend. The primary full-protein panel includes Barnase, the RNA-binding domain of E. coli Rho factor (1A63), and Stichodactyla helianthus neurotoxin I (1SH1). All guarded p4 repeats converged with finite energies, sampled direct-operator residuals below 8e-3, and one plan/cache build. Additional public proteins, Ubiquitin (1UBQ), Chymotrypsin inhibitor 2 (2CI2), and Villin headpiece subdomain (1VII), passed the same supplement gate. A dense Barnase convergence profile compares the baseline diagonal path, block-diagonal preconditioning, and sparse near-field ILU(0), showing that ILU(0) resolves a high-coupling Krylov case that the local alternatives do not. The production Poisson-Boltzmann default remains the grid backend, and the default TABI treecode path remains degree-one.

Introduction

Implicit-solvent electrostatics is a central approximation in biomolecular modeling because it replaces explicit solvent degrees of freedom with a continuum response. The Poisson-Boltzmann equation is one of the most established forms of this approximation: it combines a low-dielectric molecular interior, a high-dielectric solvent exterior, fixed molecular charges, and mobile ions. Classical APBS/DelPhi-style grid solvers are robust and widely used, but they carry a volumetric cost and represent a sharp molecular dielectric interface through a finite grid. Increasing molecular size, changing salt conditions, or refining the dielectric interface can quickly make the grid the dominant computational object.

Boundary-integral Poisson-Boltzmann methods attack the same physical problem from a surface perspective. By reformulating the linearized PB equation as coupled integral equations on the

molecular surface, they reduce the unknown set from a three-dimensional volume to a triangulated interface. This surface reduction is attractive for repeated protein calculations and for workflows in which surface discretization is the natural resolution parameter. It also creates a difficult numerical problem: the surface operator is dense, contains derivative-sensitive kernels, and includes near-singular panel interactions that cannot be treated as ordinary far-field particle sums.

Fast multipole methods provide the natural acceleration idea for such dense operators [2]. Earlier TABI/treecode and FAGBI-style work established that boundary-integral PB can be accelerated without abandoning the surface formulation. The Cartesian boundary-integral PB direction developed by Chen, Tausch, and Geng is the closest methodological reference point for TITAN's analytical FMM/M2L effort [1]. TITAN follows that broad motivation but targets a different integration setting. Rather than presenting a standalone Galerkin code, TITAN embeds an explicit guarded high-order path inside an existing TABI workflow, with production defaults left unchanged. The focus is therefore not only the analytical expansion, but also the engineering required to make the accelerated operator repeatable, auditable, and usable inside GMRES.

The second central issue is preconditioning. A boundary-integral acceleration that makes each matrix-vector product faster is not enough if the Krylov iteration count grows or the solver stalls. During TITAN development, local diagonal and local block-diagonal preconditioners repeatedly failed to capture strong coupling across tree leaves on dense molecular surfaces. Increasing local block size improved some rows but did not solve the fundamental graph problem. The final method therefore constructs a sparse near-field graph directly from FMM near-source relationships and applies block ILU(0) over coupled 2 x 2 panel blocks. This is the main numerical distinction of the present guarded path.

The claim in this manuscript is intentionally scoped. Guarded p4 is not promoted to TITAN's production default, and it is not selected automatically by TABI treecode. Instead, we document a repeatable advanced explicit path. The evidence chain asks whether the high-order operator agrees with sampled direct references, whether the preconditioner resolves a dense high-coupling case, and whether the workflow repeats on public full-protein fixtures. This is a narrower claim than a full replacement for grid PB, but it is stronger than a prototype-only demonstration.

Methods

TITAN solves a coupled boundary-integral Poisson-Boltzmann system over triangulated molecular surfaces. Each panel carries the unknown vector u_i = (phi_i, partial_n phi_i)^T, where phi is the surface potential and partial_n phi is its normal derivative. The JCP-style coupled operator contains four analytical interaction channels, denoted here as K1 through K4. These channels couple the potential and normal-derivative rows and are applied with the same production sign convention in direct, degree-one accelerated, and guarded p4 paths. GMRES is used for the nonsymmetric coupled linear system [3].

Near interactions are evaluated with the production analytic-singular kernel. This choice matters because self and adjacent-panel terms can dominate the error if they are approximated by ordinary centroid interactions. Duffy-adaptive quadrature remains available as an explicit diagnostic option [5], but it is not selected automatically in the guarded p4 workflow. The manuscript-facing policy therefore separates near-field correctness from far-field acceleration: near terms use the shared analytic-singular path, while far terms are controlled by high-order expansion, refinement, and direct fallback.

The guarded p4 far operator uses fourth-order Cartesian source moments and target-local M2L/L2P evaluation. The accepted far-cell contribution is not treated as a black-box scalar approximation. Instead, the four coupled channels are assembled from analytical Green-function jets and their source/target derivatives. The derivative channels are especially sensitive because normal derivatives and dielectric contrast multiply truncation error; a small far-field error in a scalar Green expansion can be amplified when it enters K2, K3, or K4. TITAN therefore treats the p4 expansion as a guarded candidate rather than as an unconditional acceptance rule. To keep memory bounded, TITAN stores two double-precision degree-9 Green jets per far pair, one for the Coulomb part and one for the screened part. Source-normal derivatives needed by K2 and K4 are reconstructed during contraction rather than stored as separate materialized jets.

The static refinement plan is built before GMRES and reused throughout the solve. Deterministic operator probes identify risky far cells, and a strict geometric separation guard catches outliers that an RMS tail indicator can miss. In simplified notation, the far-cell acceptor compares a p3/p4 tail indicator against the coupled output signal and then applies a geometry guard to cells whose box separation is too small relative to their radius. If either test flags the pair, the source

cell is split; if it is already a source leaf, the interaction is demoted to direct fallback. The final guarded policy uses Taylor order 4, opening theta 0.6, static tail tolerance 0.002, strict-geometry MAC theta 0.5, compressed Green-jet cache, packed AVX2 direct fallback, analytic-singular near integration, and sparse near-field ILU(0). This policy is intentionally conservative: a far cell must be both algebraically and geometrically safe before it is accepted.

The direct fallback path is also structured for reuse. The refinement plan produces deterministic target/source tasks. Non-self fallback pairs use a packed AVX2 primitive over four target lanes and source tiles arranged in cache-friendly order, while self terms are routed through the exact analytic near kernel and masked out of vector lanes to prevent double counting. This hardware-aware layout reduces branch pressure in the near-field path and keeps the high-volume fallback work close to the L1/L2 cache access pattern expected by modern CPUs. It was necessary because accuracy recovery on full 1A63 required a geometry guard that increased fallback work. Without packed fallback, the recovered p4 operator would be accurate but too expensive for guarded GMRES.

The sparse preconditioner is assembled from the FMM near-source graph. For every target panel i, the graph contains the self edge and a deterministic set of nearby source panels, capped at 64 neighbors in the guarded configuration. On each graph edge, TITAN stores a 2 x 2 block approximation to the coupled local operator: A_ij = [[delta_ij c_phi - K2_ij, -K1_ij], [-K4_ij, delta_ij c_phin - K3_ij]]. The matrix is stored in block sparse-row form. ILU(0) keeps the original graph sparsity, applying the no-fill update L_ik = A_ik U_kk^-1 and U_ij <- U_ij - L_ik U_kj only when the block (i,j) already exists [4]. The apply phase performs deterministic block forward and backward substitution.

Figure 1 summarizes the algorithmic split between far-field Cartesian p4 M2L/L2P, packed direct fallback, and the near-source graph used for sparse ILU(0). Source panels are compressed into Cartesian moments, accepted far cells are translated into target-local expansions, and near or geometrically risky interactions are routed through the shared direct P2P path.

Protein structures are prepared with PDB2PQR/APBS-family tooling [6], [7] and public structure identifiers from the RCSB Protein Data Bank [8]. The primary full-protein panel contains Barnase, the RNA-binding domain of E. coli Rho factor (1A63), and Stichodactyla helianthus neurotoxin I (1SH1). The supplementary public-protein expansion contains Ubiquitin (1UBQ),

Chymotrypsin inhibitor 2 (2CI2), and Villin headpiece subdomain (1VII). Manuscript-facing rows are full-protein fixtures, not atom-sampled fixtures.

The reference policy is matched to computational feasibility. For analytic and bounded development tests, direct TABI can be used as a full reference. For the full-protein guarded p4 panel, full direct TABI reaction energies are not practical for the largest rows, so verification uses sampled direct-operator RMS and sampled direct true residuals. Grid PB remains the production default and is useful runtime context, but it is not treated as the oracle for the high-order boundary-integral path.

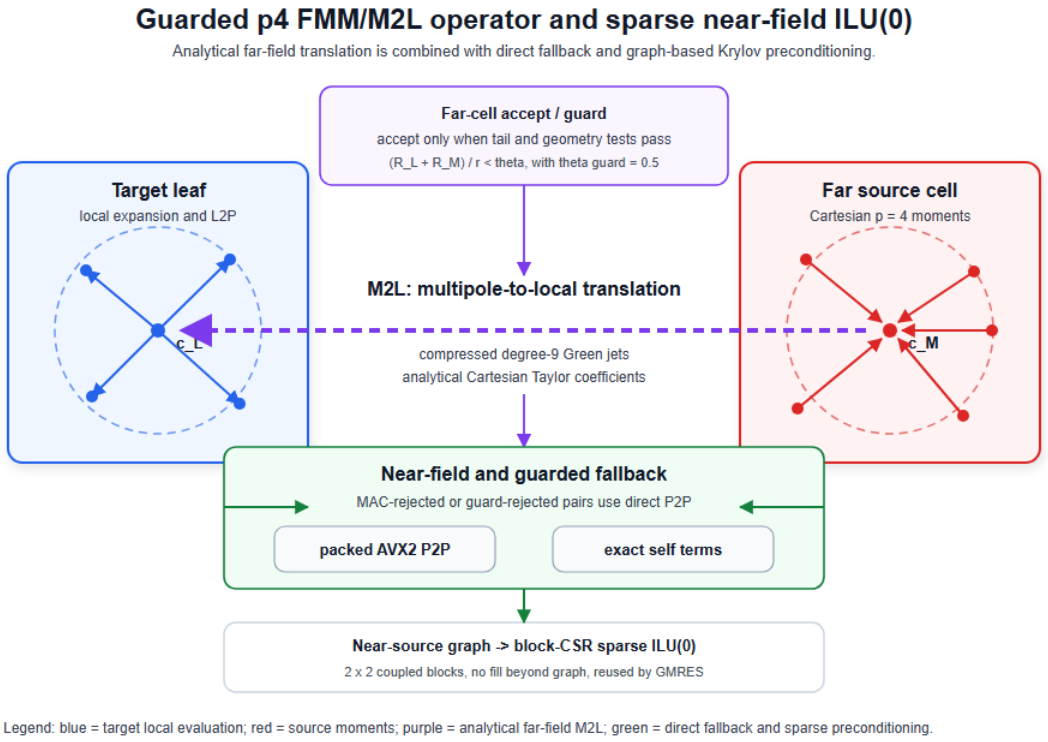


Figure 1. Guarded p4 FMM/M2L boundary-integral operator. Source panels are compressed into Cartesian p4 moments, translated to target-cell locals by analytical M2L, and evaluated by L2P. Near-field and geometrically risky interactions are routed to packed direct P2P fallback, while the same near-source graph defines the sparse block ILU(0) preconditioner used by GMRES.

Analytical Validation Protocol

Before the full-protein guarded p4 rows were accepted, TITAN used controlled analytic systems to separate surface-discretization error from acceleration error. Centered Born and off-center Kirkwood-style tests were used to check whether direct TABI with analytic-singular near

integration converged toward known spherical references as the surface was refined. This step matters because an accelerated operator cannot be meaningfully validated if the underlying surface discretization is already outside the target accuracy range.

The high-resolution direct TABI checks showed that the remaining analytic error could be reduced below the manuscript threshold on the controlled sphere cases. Those tests do not prove full-protein energy accuracy by themselves, because protein surfaces are more irregular and do not have closed-form reaction-energy oracles. They do establish that the panel basis, normals, diagonal terms, and analytic-singular near path are not the dominant explanation for the later full-protein operator gaps. This allowed the later debugging to focus on far-field truncation, refinement, and preconditioning rather than on the direct near-field kernel.

The operator attribution sequence then separated near and far contributions on full proteins. The key observation was that a nominal fourth-order configuration could still fail on 1A63 if the refinement plan was driven only by an averaged tail estimate. The outlier was not a sign-convention error, a near-kernel regression, or a cache lifecycle bug. It was a far-cell acceptance problem: a small number of geometrically risky accepted cells could dominate the sampled direct residual. The final strict-geometry guard is the direct response to that attribution.

Lifecycle And Reproducibility Controls

The guarded p4 path is prepared before GMRES. The context builds the FMM geometry, static refinement plan, compressed Green-jet cache, packed fallback worklist, and near-field ILU(0) preconditioner once. During Krylov iteration, only source moments and density-dependent contractions are updated. This lifecycle is part of the numerical contract, not only a performance detail. If the plan or cache were rebuilt inside GMRES, repeated runs could become difficult to interpret and setup cost would contaminate matvec timing.

Every manuscript-facing guarded p4 row therefore records plan/cache build counts. The accepted rows have build count one. They also record sampled direct-operator residuals after convergence, so the reported solver residual is not the only accuracy measure. Internal GMRES residual can be small even when the accelerated operator differs from the direct operator; the full 1A63 diagnostic exposed exactly that failure before the geometry guard was added. The final verification requires both internal convergence and sampled direct true residual to pass.

The implementation uses deterministic ordering for refinement, cache layout, fallback tasks, and panel sampling. Parallelism is allowed, but worklists are sorted by target and source identifiers, and each worker owns disjoint output panels. This determinism is why repeat energy spread is a meaningful package metric. A repeatability table with zero spread at reported precision is not a substitute for independent validation, but it does show that the guarded path is not depending on race-sensitive accumulation or accidental cache state.

Results And Discussion

The primary guarded p4 panel passed all repeatability gates. Barnase, 1A63, and 1SH1 each converged with finite energies, sampled direct verification below 8e-3, and one plan/cache build per solve. The largest sampled direct residual in the primary panel occurred for 1A63, approximately 2.36e-3, still well inside the gate. The energy spread across repeats was zero in the recorded package, indicating deterministic solver behavior at the reported precision.

The full six-protein performance ladder provides a clearer algorithmic picture than a single guarded p4 row. Classical direct TABI remains the most transparent reference path, and this work builds on the solid TABI and FAGBI foundation rather than treating earlier formulations as defective. In the matched full-protein ladder, however, direct TABI was not a practical production path under the present time budgets: 2 rows reached the direct timeout and 4 additional rows stopped with residual/status gaps. This is consistent with the dense-operator bottleneck already identified in the original TABI literature.

The degree-one FMM rows separate matvec acceleration from linear-solver robustness. Diagonal and local block preconditioning reduced the cost of individual operator applications, but these rows still frequently reached the iteration budget or failed the requested residual status. Sparse near-field ILU(0) changed the Krylov behavior: every degree-one FMM + ILU(0) row converged in approximately 128 iterations. That result isolates the value of graph-based preconditioning. At the same time, the largest degree-one audit remained 9.671e-01, showing that convergence of the low-order accelerated operator is not the same as agreement with the sampled direct operator.

The final guarded p4 path combines the two requirements. Across Barnase, 1A63, 1SH1, 1UBQ, 2CI2, and 1VII, all guarded p4 + sparse ILU(0) rows converged in 128 iterations, with maximum sampled audit 2.534e-03, maximum sampled direct true residual 2.535e-03, and peak resident

memory 369.6 MB. Thus, the high-order guarded operator restores direct-operator agreement while preserving the convergence benefit of sparse near-field ILU(0). The performance ladder table and central runtime figure summarize this progression from dense direct TABI, to accelerated but weakly preconditioned FMM, to convergent degree-one FMM + ILU(0), and finally to guarded p4 + ILU(0).

The supplementary public-protein expansion broadens the evidence without changing the main claim. Ubiquitin, Chymotrypsin inhibitor 2, and Villin headpiece subdomain passed the same guarded p4 repeatability gate. These systems are useful because they are public, compact, and prepared as full proteins. They are treated as supplement evidence rather than as a replacement for the locked primary panel.

The Barnase convergence profile provides the clearest preconditioner contrast. Figure 4 compares the baseline diagonal path, local block-diagonal preconditioning, and sparse near-field ILU(0) on the same full-protein degree-one FMM operator. The diagonal and block-diagonal rows both reached the 720-iteration budget, with sampled true residuals of 4.420e-02 and 4.786e-02, respectively. Sparse near-field ILU(0), by contrast, converged in 128 iterations with sampled true residual 1.319e-12. This profile is the strongest evidence that the decisive improvement is not merely faster far-field evaluation, but a change in the algebraic coupling available to the Krylov solver. Sparse near-field ILU(0) uses the same near-source graph exposed by FMM and allows information to propagate across neighboring panels. The result should be interpreted narrowly: it demonstrates a hard coupling regime, not a claim that every protein requires sparse ILU(0).

Three lessons follow. First, far-field accuracy for derivative-rich boundary-integral PB operators is not controlled by expansion order alone. A fourth-order Cartesian expansion was necessary but not sufficient; full 1A63 exposed outliers that passed a weaker averaged refinement policy. The final strict geometry guard recovered the sampled direct residual while keeping fallback work within budget. Second, preconditioning must follow the coupling graph. Local diagonal and local block-diagonal methods are simple and memory-bounded, but they can miss strong cross-leaf interactions. Third, hardware-aware near-field work matters: once risky far cells are demoted, packed fallback becomes part of the algorithm rather than a slow exception path. Fourth, evidence discipline matters: the guarded p4 path is documented as a repeatable advanced explicit

TABI treecode option, not as an automatic replacement for the production grid backend or the default degree-one TABI treecode path.

Compared with the FAGBI reference direction [1], this work emphasizes integration into an existing molecular electrostatics codebase. The algorithmic components are not only analytical FMM/M2L expansions, but also static refinement, compressed jet caching, vectorized fallback, lifecycle control inside GMRES, and graph-based sparse preconditioning. The manuscript should therefore be read as a continuation of the TABI/FAGBI development line: it preserves the boundary-integral motivation while adding guarded high-order evaluation, sparse near-field preconditioning, and repeatable full-protein validation.

Two design choices are worth making explicit for reviewers. First, TITAN uses Cartesian Taylor moments rather than spherical harmonics in this guarded implementation. Although spherical-harmonic translations have attractive asymptotic structure, the fixed $p = 4$ regime used here is dominated by constants, memory layout, and derivative-channel bookkeeping. Cartesian jets keep the K1-K4 construction local, unrolled, and compatible with the packed fallback and compressed-cache lifecycle. Second, grid and boundary-integral energies should not be treated as interchangeable oracles. Grid PB smooths a dielectric interface over a volume mesh, whereas BIE evaluates a sharp molecular boundary. The manuscript therefore uses sampled direct-operator residuals as the operator agreement test for large proteins and treats grid rows as production/default context rather than as the mathematical reference for the guarded surface operator.

Limitations

Several limitations are deliberate. Full direct TABI reaction energies are not used as the oracle for large full-protein rows; sampled direct-operator residuals are the feasible verification mechanism. Duffy-adaptive near integration remains explicit-only and is not automatically selected. The Barnase preconditioner convergence profile is framed as dense/high-coupling evidence rather than a broad claim over all proteins. The supplementary fixtures broaden the evidence but do not replace the primary locked panel.

The manuscript also does not claim a general production promotion. Guarded p4 remains an advanced explicit TABI treecode option, and grid remains TITAN's production Poisson-

Boltzmann default. Future work should evaluate larger public protein panels, compare against independent PB packages under matched surface preparation, and test whether the guarded p4 policy remains optimal across ionic strengths, dielectric settings, and finer surface discretizations.

Reproducibility And Data Availability

The manuscript package is designed to be auditable from tracked artifacts rather than from undocumented local state. The tables, figure inputs, reference list, BibTeX file, claim trace, citation-gap closure, and rendered page checks are written into the paper package. The public-protein fixtures are identified by biological name and PDB identifier, and the preparation workflow records the PDB2PQR/APBS-family provenance used for the supplementary systems. The package therefore separates numerical evidence from manuscript formatting: rebuilding the document does not rerun the solver, and rerunning the solver gates produces new matrix artifacts that can be compared with the frozen paper package.

All large-protein accuracy statements in the manuscript are tied to sampled direct-operator verification rather than to unreported full direct solves. This is a limitation, but it is also a reproducibility choice. The sampled audit rows are deterministic, and the manuscript records the policy boundary explicitly so that future work can replace the sampled reference with full direct energy checks when practical. The current package should therefore be read as a reproducible evidence-backed draft, not as a final journal submission with independent external-code validation.

Conclusion

TITAN now contains a repeatable guarded fourth-order analytical FMM/M2L boundary-integral PB path with sparse near-field ILU(0) preconditioning. On the primary full-protein panel and supplementary public fixtures, the explicit guarded p4 workflow converges, passes sampled direct-operator verification, and preserves a clear policy boundary: it is a documented advanced option, not the production default. The evidence supports a paper-facing claim centered on analytical acceleration, deterministic far-field guarding, and graph-based sparse preconditioning for dense molecular-surface PB solves.

Algorithmic Performance Ladder



Figure 2. Full-protein performance ladder. Colors identify backend stages from direct TABI through guarded p4 + sparse ILU(0); each cell reports status and wall time. The guarded p4 + sparse ILU(0) rows passed sampled direct-operator verification on all six full proteins.

Table 1. Full-protein benchmark systems used in the performance ladder.

Protein	PDB/source	Atoms	Panels	Role
Barnase	PyDelPhi benchmark	1700	14942	Primary
RNA-binding domain of E. coli Rho factor	1A63	2065	17250	Primary
Stichodactyla helianthus neurotoxin I	1SH1	696	6673	Primary
Ubiquitin	1UBQ	1231	11242	Supplement
Chymotrypsin inhibitor 2	2CI2	1076	9889	Supplement
Villin headpiece subdomain	1VII	596	5032	Supplement

Table 2. Full-protein algorithmic ladder across six matched protein fixtures.

Stage	Outcome	Median wall	Median iter.	Operator audit	Interpretation
Classical direct TABI	0/6 ok; 4 failed; 2 timeout	31.1 min	576	direct reference	Valid reference path, but dense direct matvecs dominate runtime at full-protein panel counts.
Degree-one FMM + diagonal	0/6 ok; 5 failed; 1 timeout	944.5 s	680	median 1.207e-02	Matvec acceleration alone was not enough for robust Krylov convergence.
Degree-one FMM + local block	0/6 ok; 6 failed; 0 timeout	700.8 s	720	median 1.258e-02	Matvec acceleration alone was not enough for robust Krylov convergence.
Degree-one FMM + sparse ILU(0)	6/6 ok; 0 failed; 0 timeout	133.3 s	128	median 4.052e-02; max 9.671e-01	Sparse ILU(0) made GMRES robust, but the low-order far operator still failed direct-operator audits.
Guarded p4 FMM + sparse ILU(0)	6/6 ok; 0 failed; 0 timeout	140.6 s	128	median 1.121e-05; max 2.534e-03	High-order guarded FMM/M2L retained ILU(0) convergence and restored sampled direct-operator agreement.

Table 3. Guarded p4 + sparse ILU(0) full-protein rows from the performance ladder.

Protein	PDB/source	Panels	Iter.	Internal residual	Audit RMS	Direct residual	Wall	RSS MB
Barnase	PyDelPhi benchmark	14942	128	8.579e-13	7.647e-05	7.647e-05	245.0 s	172.5
RNA-binding domain of E. coli Rho factor	1A63	17250	128	1.337e-11	2.534e-03	2.535e-03	312.6 s	369.6
Stichodactyla helianthus neurotoxin I	1SH1	6673	128	1.164e-14	5.519e-06	5.519e-06	57.0 s	63.1
Ubiquitin	1UBQ	11242	128	2.719e-13	1.440e-05	1.440e-05	154.9 s	144.6
Chymotrypsin inhibitor 2	2CI2	9889	128	5.964e-14	8.022e-06	8.022e-06	126.3 s	138.1
Villin headpiece subdomain	1VII	5032	128	7.353e-13	5.357e-06	5.357e-06	37.4 s	54.7

Primary Protein Panel

Table 4. Guarded p4 repeatability on the locked full-protein panel.

Protein	PDB/source	Panels	Energy	Audit max	Direct residual max	Energy spread	p4/deg1
Barnase	PyDelPhi benchmark	14942	-21030.83	6.383e-05	6.383e-05	0.000	0.450
RNA-binding domain of E. coli Rho factor	1A63	17250	-247716.08	2.357e-03	2.357e-03	0.000	0.674
Stichodactyla helianthus neurotoxin I	1SH1	6673	-4362.07	4.894e-06	4.894e-06	0.000	0.383

Supplementary Repeatability Context

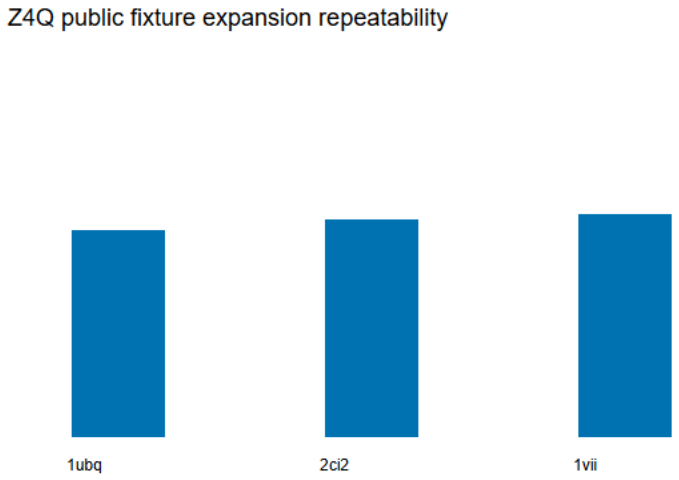


Figure 3. Supplementary public-protein guarded p4 repeatability for 1UBQ, 2CI2, and 1VII.

Preconditioner Convergence Profile

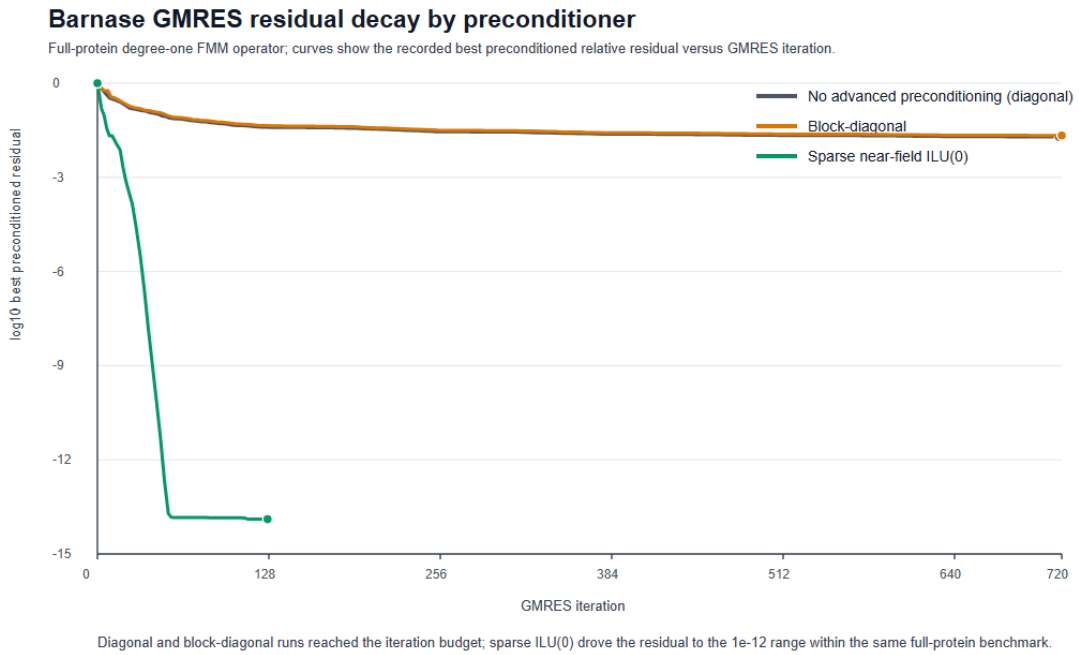


Figure 4. Barnase GMRES residual decay under three preconditioning choices on the same full-protein degree-one FMM operator. The baseline diagonal path and local block-diagonal preconditioner reached the iteration budget, whereas sparse near-field ILU(0) drove the preconditioned residual down to the 1e-12 range and produced a converged solve.

Supplementary Public-Protein Expansion

Table S1. Full public-protein supplement rows prepared without atom sampling.

Protein	PDB	Atoms	Panels	Energy	Audit max	Direct residual max	Energy spread
Ubiquitin	1UBQ	1231	11242	-4038.46	1.468e-05	1.468e-05	0.000
Villin headpiece subdomain	1VII	596	5032	-2355.70	5.871e-06	5.871e-06	0.000
Chymotrypsin inhibitor 2	2CI2	1076	9889	-4891.15	7.983e-06	7.983e-06	0.000

References

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